

2011 HSC Notes from the Marking Centre – Chemistry

Contents

[Introduction](#)

[Section I – Core](#)

[Section II – Options](#)

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Chemistry. It contains comments on candidate responses to the 2011 Higher School Certificate examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

This document should be read along with the relevant syllabus, the 2011 Higher School Certificate examination, the marking guidelines and other support documents developed by the Board of Studies to assist in the teaching and learning of Chemistry.

General comments

Teachers and candidates should be aware that examiners may ask questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course including the prescribed focus areas. It is important to understand that the Preliminary course is assumed knowledge for the HSC course.

Candidates need to be aware that the marks allocated to the question and the answer space (where this is provided on the examination paper) are guides to the length of the required response. A longer response will not in itself lead to higher marks. Writing in excess of the space allocated may reduce the time available for answering other questions.

Candidates need to be familiar with the Board's [Glossary of Key Words](#), which contains some terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the key words from the glossary. Questions such as 'how?', 'why?' or 'to what extent?' may be asked or verbs may be used that are not included in the glossary, such as 'design', 'translate' or 'list'.

Teachers and candidates are reminded that mandatory skills content in Module 9.1 are examinable in both the Core and Option questions.

Candidates should use examination time to analyse the question, plan their responses carefully then work within that framework to produce clear, logical and concise responses. The response may include the use of dot points, diagrams and/or tables, and planning the response will help avoid internal contradictions. Holistic responses need to be logical, well constructed and relevant to the questions asked.

In better responses, candidates:

- set out all working for numerical questions and did not round off until the final answer
- thought carefully about the units to be used and the quantities to be substituted into formulae
- did not repeat the question as part of the response
- looked at the structure of the whole question and noted that in some questions the parts followed on from each other, eg responses in part (a) led to the required response in part (b)
- used appropriate equipment, eg pencils and a ruler to draw diagrams
- accurately transcribed and used values from the periodic table and data sheet for calculations
- included balanced chemical equations where appropriate with the correct states of all species shown
- were guided by the space provided on the paper regarding the length of the response necessary.

Candidates are strongly advised to answer the question relating to the Option they studied in class.

Section I – Core

Part B

Question 21

In better responses, the polarity of the ethanol molecule was correctly identified and linked to the appropriate intermolecular forces. These

responses included examples of the use of ethanol as a solvent for polar and non-polar substances, as well as annotated diagrams. They included correct annotation of the dipole on the hydroxyl group. In some responses, the small size of ethanol relevant to its role as a solvent was correctly understood.

In weaker responses, candidates confused the structure of ethanol with those of ethene or ethanoic acid or labelled the dipole incorrectly. Some responses incorrectly showed a positive pole on one end of the molecule and a negative pole at the other. In weaker responses, candidates neglected to show hydrogen atoms, lined them up incorrectly or confused the position of dipoles. They confused hydroxyl group with hydroxide ion, alkyl group with alkane.

Candidates are reminded that a structural formula should show every atom and every bond.

Question 22

- a. In better responses, candidates included at least two correct balanced equations relevant to the depletion of ozone. Candidates are reminded that the chlorine atom responsible is a free radical and not a chloride ion.
- b. Better responses identified a method used to detect stratospheric ozone and provided a description of the method. Some responses correctly identified the monitoring element in the question.

Question 23

- a. In better responses, candidates recognised that Cn-278 is unstable because of the size/mass of the nucleus. In weaker responses, candidates referred to the proton/neutron ratio or confused instability with being transuranic. Some responses displayed confusion between stability and reactivity.
- b. In better responses, candidates described the process whereby transuranic isotopes are synthesised in nuclear reactors or particle accelerators. Some candidates included examples of relevant nuclear equations. In weaker responses, the process of transuranic isotope formation was not correctly paired with the location of the process. Many responses were vague on the nature of the particles interacting with the large nuclei and the speed or method of interaction.

Question 24

- a. This question was generally well done, however a number of candidates were unable to provide a net ionic equation for the process. Better responses included a correctly balanced net ionic equation for the process plus a standard cell potential for the process described in their equation. Weaker responses demonstrated confusion in the direction of the electron flow or provided a standard cell potential which was inconsistent with the included equation (eg Cu^{2+} was used in the equation but the standard potential for Cu^{+} was calculated).
- b.
 - i. The best responses were clearly set out and included working. Most candidates calculated the number of moles of copper gained. Most responses related the number of moles lost by the nickel electrode to the number of moles gained by the copper electrode but weaker responses did not convert this to mass of nickel lost and/or subtract this from the original mass of the nickel electrode. In some responses, the calculated mass of nickel was added rather than subtracted, suggesting that there was some confusion about the site of oxidation. In others, the answer was rounded off too early in the process leading to an unacceptably large deviation in the final answer. In the weakest responses the mass of nickel lost was equated directly to the mass of copper gained or the percentage of mass gained by copper electrode was calculated and equated directly to the percentage of mass lost by nickel electrode.
 - ii. Many responses stopped with the calculation of the original concentration of the solution. In others the additional concentration was calculated but was not added to the original. A common mistake was to use the mass of the nickel electrode calculated in 24 bi to calculate the concentration of the solution. In a considerable number of responses, candidates inappropriately used the dilution equation $C_1V_1 = C_2V_2$ in attempting to calculate the final concentration.

Question 25

In better responses, candidates identified the conjugate acid base pair of $\text{H}_2\text{PO}_4(\text{aq})/\text{HPO}_4^{2-}(\text{aq})$ as a buffer system and, using appropriate equations, clearly explained how the buffer system works when additional acid and base are added. In weaker responses, candidates did not include relevant balanced equations and displayed some confusion about how the system changes to counteract addition of acid and/or base.

Question 26

- a. Better responses included the identification of the hygroscopic/deliquescent nature of sodium hydroxide and the calculation of the standardised and theoretical concentration of the NaOH solution. In most responses, the hygroscopic/deliquescent nature of NaOH was indicated but not that the standardised value was lower than the theoretical value. In weaker responses, it was implied that the NaOH had reacted with water rather than absorbed water.
- b. Better responses clearly set out all working, did not include the outlier when determining the average titre volume, identified the triprotic nature of citric acid and used it in their calculation, used the standardised concentration value of the NaOH solution rather than the theoretical value based on mass, recognised and used the appropriate dilution factor to determine the original citric acid concentration and did not round off figures in the early stages of the calculation.

Question 27

In better responses, candidates clearly demonstrated a thorough understanding of the relationship between the properties and uses of both biopolymers and how they affected the environment, linking these to the blog. A significant number of responses did not include an assessment that linked the properties of the polymers to the blog. They included an assessment, but did not address both uses and properties of the polymers. In most responses properties and uses of polystyrene were identified. Most candidates used the biopolymer 'Biopol', identified its uses and that it is biodegradable.

Question 28

In better responses, candidates distinguished two tests based on a chemical or physical principle. This principle was then considered when demonstrating an understanding of the test required. First-hand experience in doing the water testing was evident through a cohesive and logical outline of the method used and evidence of how the outcome of the experiment was observed. In weaker responses, the connection between the principle and the test was not evident and steps in their procedure were not included.

Question 29

- a. In better responses, candidates demonstrated a working knowledge of both the Arrhenius definition of an acid and the Bronsted-Lowry definition of an acid and a base. They showed the importance of the role of the solvent in both definitions and the application to current use of acids and bases in the laboratory. In weaker responses, candidates were not clear in their definitions – especially with the concept of proton transfer in the Bronsted-Lowry definition – and did not include evidence of support statements.
- b. Better responses demonstrated, in words or with an equation, that the net reaction between a strong acid and a strong base will always be the same. Weaker responses did not distinguish the importance of the strength of the acid and base in always giving the same heat of reaction. The complete ionisation of a strong acid and complete dissociation of a strong base was not recognised.

Question 30

In better responses, candidates recognised that Step 1 and Step 2 were equilibrium reactions and were able to apply Le Chateliers Principle to explain how concentration, temperature and pressure affected yield with specific reference to the exothermic nature of the reactions and the mole ratio of gases. Responses demonstrated an understanding of the need to use some compromise conditions to balance yield and reaction rate. Candidates recognised that Step 3 was a reaction that went to completion and consequently that reaction conditions could only affect the reaction rate and not the yield.

In better responses, candidates demonstrated an understanding of collision theory, activation energy and the use of catalysts, temperature and pressure to increase reaction rate. They were logical, coherent and recognised that some of the same reaction conditions that affected yield could be applied to both Step 1 and Step 2, and factors that affect reaction rate could be applied to all 3 steps.

In weaker responses, candidates simply identified conditions that affected yield, but did not explain them. Some candidates simply addressed either yield or reaction rate. In weaker responses, candidates incorrectly identified heat as a catalyst. Many weaker responses addressed Step 3 as if it were an equilibrium reaction.

Question 31

In better responses, candidates included valid sources of the contamination occurring on both Tuesday and Thursday. Candidates also demonstrated knowledge that the microbes required oxygen to decompose the organic waste leading to the increased BOD and subsequently the reduction in dissolved oxygen. Weaker responses provided a valid source, but did not indicate how the source could enter the river.

Question 32

- a. In correct responses, candidates indicated that barium sulfate was white and insoluble which would make it a suitable background to observe the colour of the indicator.
- b. In better responses, candidates correctly identified the pH range of the soil as being between 4–4.5. They identified that Hydrangea Blue was the only plant that could grow optimally under these conditions. Other responses included that Hydrangea Blue would grow optimally but did not indicate the pH of the soil. Some responses included an incorrect pH range of the soil.
- c. Better responses indicated that solutions to be tested needed to cover the entire pH range of 1–14. These responses stated that the indicator needed to be added to the tested solutions and that the colour of the indicator observed. In some responses, candidates indicated that acidic, basic and neutral solutions were tested and that the colour of the indicator for each was observed. Poorer responses gave details of a method used to prepare a natural indicator but did not provide sufficient details of testing the indicator.

Section II – Options

Question 33 – Industrial Chemistry

- a. Many candidates did not recognise the equipment shown in the stimulus as a dessicator, so did not postulate the identity of the beaker of liquid in terms of dehydration. Candidates often deduced that the sealed container was designed to model aspects of equilibrium. Candidates who could deduce the purpose of the experiment were often able to identify the liquid as sulfuric acid and as a dehydrating agent.

In the better responses, candidates correctly identified the liquid as concentrated sulfuric acid. Candidates who were able to identify sulfuric acid acting in the sealed container often incorrectly explained its other use as an oxidising agent. Few candidates supported their answer with a coherent chemical equation to show the process of dehydration. Some candidates demonstrated all of the key

criteria, including correct colour change and provide a supporting balanced chemical equation.

- b. i. Many candidates correctly stated that the carbon dioxide was not of any environmental concern because it was recycled or reused within or during the Solvay process. Weaker responses did not always clearly state how or where it was reused, often stating that it was used for 'other processes'. Some correct responses were clouded with extra information on the consumption of any released carbon dioxide by natural processes such as photosynthesis. Responses were often supported with at least one balanced equation for both the production and consumption of carbon dioxide. Many responses attempted to provide more than one equation for each process. Better responses included the Solvay process, supported with a consumption equation, but did not include an equation for the production of carbon dioxide.
- ii. In better responses, candidates often provided a correctly balanced equation for the net Solvay process as a means of showing the mole ratio for NaCl to CaCl₂ as being 2:1 where a balanced equation was not provided the calculation as if it were based on a 1:1 ratio or the use of a correct ratio had to be deduced via their sequence of calculations. Many candidates correctly processed the data and achieved the correct answer including appropriate units of grams, kg or tonnes. Some responses indicated that candidates converted 1 tonne to 1 000 000 grams. Subsequent calculations were therefore incorrect.
- c. i. Many candidates did not note that the NaCl solution was only 0.05 M and that the electrolysis would only be that of water and not of the NaCl. In better responses, candidates provided both correct oxidation and reduction half-equations for water and nominated the anode and the cathode as the + and – respectively. Many responses included a correct answer for the cathode but incorrectly provided a half-equation for the oxidation of chloride at the anode with associated explanations for the observations listed in the table. Stronger responses included both correct half-equations as well as correct accounts for all four of the observations in the table. The litmus colours were often well explained in terms of the H⁺ and OH⁻ production at each electrode. Poorer responses stated red due to acid and blue due to base. The bubbles of gas evident at each electrode were often ignored, poorly explained or possibly assumed as self evident if shown in the half-equations. Transcription of the half-equations from the reduction potential table was generally well done, including the reversal of one equation to show oxidation. Double arrows were often left in the answers. The labelling of the positive electrode as the anode and the negative as the cathode during electrolysis was consistent in the stronger responses.
- In weaker responses, candidates confused the labelling and also the processes of oxidation or reduction occurring for each electrode. The provision of incorrect oxidation or reduction half-equations in support of their anode/cathode labels highlighted this.
- ii. Statements such as Electrolytic cells – Electrical energy → Chemical energy OR
Galvanic cells – Chemical energy → Electrical energy
when provided, earned candidates the marks.

In better responses, candidates provided descriptions of the energy considerations for each process before stating the required answer. Many of the stronger responses provided excellent answers demonstrating comprehensive understanding of a comparison of the two cells in terms of E⁰ potentials being negative for electrolytic and positive for galvanic, the voltage requirements or production, but were not explicit about all of the energy inputs/outputs in terms of electrical and chemical.

Candidates who explained the energy inputs/outputs of one cell, but did not include a comparison with the other cell were unable to earn the marks. Some candidates neglected to include energy considerations in their answers and did not earn the marks.

- d. i. Stronger responses described a physical model of equilibrium that could be performed in the classroom, while some weaker responses described an actual chemical equilibrium reaction. Better responses related the main features of their model to a system at equilibrium.
- ii. Better responses addressed both the limitations of models and factors that promoted validity. Poorer responses displayed confusion between the validity and the reliability of data, and there was some evidence of rote-learned definitions of validity that could not be applied to the question.
- e. Better responses included a description of processes and a discussion of the impacts of environmental issues. Weaker responses did not address different processes; did not relate the environmental issue to the impact on society, nor how the impact could be mitigated.

Question 34 – Shipwrecks, Corrosion and Conservation

- a. Most candidates named the material as aluminium, magnesium or zinc for X. In better responses, candidates indicated that X was a sacrificial anode in a galvanic cell and justified the choice of X by referring to the relevant reduction potentials for their named metal and iron. They used terminology such as oxidation, reduction, anode and cathode appropriately to describe the galvanic cell formed.

Weaker responses used more general terms, such as more reactive, to compare the reduction potentials. A significant proportion of the responses indicated incorrect interpretation of the term 'anticorrosion rod', using terms such as protecting the rod from corrosion or passivating metals.

- b. i. Many candidates incorrectly referred to this process as an electrolytic cell. Stronger responses identified the galvanic cell formed by

the contact of the aluminium tray and silver cutlery in sodium hydrogen carbonate electrolyte solution and used the table of standard potentials to show that aluminium would provide electrons to the cutlery in order to reduce silver ions back to metallic silver. Only a small number of candidates provided two half equations and correctly added them to give an overall equation. Weaker responses showed that aluminium was oxidised and silver was reduced, but there were no supporting E° values or equations.

- ii. Most candidates did not identify an advantage that demonstrated chemical knowledge. Better responses indicated that the galvanic process would restore the silver by reducing the silver ions in the tarnish back to metallic silver.
- c. i. Most candidates drew an electrolytic cell labelling electrodes, power supply and electrolyte. Better responses clearly indicated the positive anode, negative cathode and electron flow. However, the diagrams were generally poorly labelled. Candidates are encouraged to use appropriate terminology and use arrows/lines to clarify their intent.
- ii. Fewer than half the candidates correctly identified oxygen and hydrogen as the gases produced. Better responses included the appropriate equations from the data sheet, matched correctly to the anode and cathode. In weaker responses, the equation was not matched to the correct electrode or included incorrect references to the reduction of potassium or sulfate ions.
- d. i. In better responses, candidates explained a suitable method to prepare the sample and provided a reason for the method. Better responses distinguished between sample and investigation. Many responses suggested how to improve the investigation in this section or referred to the reliability and validity.
- ii. In better responses, candidates assessed the validity of data by providing both attributes and limitations. These responses included assessing the validity in terms of control of variables. Poor responses indicated that candidates could not clearly distinguish between valid and invalid.
- e. Better responses included relevant, balanced, and correct equations. Better responses included the name of a recent maritime investigation and provided a coherent and logical value judgement. Many responses described the conditions in terms of corrosion. Poor responses diverted to the restoration and conservation of artefacts and wrecks.

Question 35 – Biochemistry of Movement

- a. In better responses, candidates correctly identified X as the Krebs Cycle or TCA cycle and articulated the difference in oxygen requirements between the Krebs Cycle and the chemical processes in the mitochondria. Some candidates had a vague idea about the oxygen requirements for chemical processes in the mitochondria but failed to identify process X correctly or specify the oxygen requirements.
- b. i. In better responses, candidates included a correct general formula for amino acids and used this formula correctly to show the addition of two amino acids to form a di-peptide with the production of a water molecule. In weaker responses, candidates used an incorrect formula or a specific amino acid and left out the production of one water molecule.
- ii. Better responses included the effect of increased temperature and pH leading to denaturation and the disruption of the secondary and tertiary structures of the protein. These responses also included the types of bonds that were broken. In weaker responses, candidates referred to denaturation without reference to the types of bonds or the particular level of structure that was affected.
- c. i. Most candidates addressed the two types of muscle cells and their properties. In better responses, candidates cited several of the properties for each type and link these properties to the use of the muscle cell, as well as extracting information from the table to aid in their response. Poorer responses failed to make more than a cursory summary of the main properties of these two muscle types.
- ii. Better responses included the energy term on the right side of the equation or noted that the reaction was endothermic. Poorer responses noted that the reaction was exothermic or had the reaction going in the reverse direction without an energy term.
- d. i. Most candidates recognised that enzymes worked in a very narrow range of temperatures and that— for humans—the optimal temperature was 37°C. In better responses, candidates noted the reaction rate being highest at the optimal temperature and the enzyme being slow or inactive at lower than optimal temperatures.
- ii. Most candidates generally addressed independent, dependent and controlled variables as well as repetition necessary for a valid experiment but were unable to specify what they were in this question. In better responses, candidates linked the variable (dependent, independent and controlled) to the validity of the experiment and formed a judgement with reasons.
- e. While most candidates showed some knowledge of the biochemical pathways involved in both carbohydrate and fat metabolism, the majority failed to link this knowledge to energy availability or to society's current understanding of nutrition. Few candidates discussed energy production and availability with the metabolism of fats or carbohydrates.

Question 36 – Chemistry of Art

- a. Most candidates identified that the cause of the different colours of the complex iron was the different oxidation state of the iron. In

better responses, candidates linked the change in oxidation state to a change in the electron configuration, hence energies associated with the d orbitals in the ions and explained the different colours in terms of different electron transitions. There was evidence of confusion in the concept of colour via absorption and some responses incorrectly explained colour due to emission of wavelengths as electrons drop to lower orbitals, rather than absorption as they move to high ones. Better responses included the concept of complementary colours and a small number included examples using the colours of the complexes in the question.

- b. i. Most candidates identified a cation and anion with the same electron configuration as argon. There was some confusion in a small number of responses between the terms cation and anion.
- ii. Most candidates explained why Group I and Group II metals only had one oxidation state in terms of the electron configuration of these elements. In better responses, candidates explained why transition metals were able to form multiple oxidation states, by relating the fact that their valence electrons are in closely spaced 's' and 'd' orbitals to their ability to lose or use a variety of electrons when forming bonds.
- c. i. Most candidates gave a partial description of how line emission data is measured. There was evidence of confusion between methods used to view absorption spectra and emission spectra, and many responses omitted an important step in the measurement of the wavelengths. Better responses included a clear method of excitation of the elements and identified a spectroscope as the instrument required to determine the wavelengths.
- ii. Many responses to this question only demonstrated partial understanding of energy level diagrams, with one or more important features of the diagram being omitted. Better responses showed diagrams including four lines, with arrows pointing downwards and the lines labelled to show which line in the spectra corresponded to each drop, and included an approximate scale.
- d. i. Most candidates described a basic method of conducting a flame test to investigate the emission colour of Group II metals. Better responses identified the need for a blue flame and the requirement that the metal be in the form of a salt or ions in a solution.
- ii. Most candidates identified a positive or negative aspect related to the validity of a flame test, but there was some evidence of confusion between the terms validity and reliability. Better responses included a number of limitations and benefits of their method and/or flame tests in general, as well as an overall assessment of the procedure based on the positives/benefits described.
- e. Many candidates identified a number of relevant spectroscopic techniques used in pigment analysis, but often the responses demonstrated only a basic understanding of the techniques. Better responses included detailed descriptions of three techniques using correct scientific terminology, the relationships between each technique and the information about pigment composition each is able to provide. A relatively small proportion of responses included examples of the type of pigment each technique could identify. In better responses, candidates showed a thorough knowledge of the type of pigments used in medieval paintings, including correct chemical formula.

Question 37 – Forensic Chemistry

- a. In better responses, candidates identified the structure as the disaccharide maltose, justifying that the open chain structure of glucose has a carbonyl group, which can be oxidised. Weaker responses identified the sugar as sucrose yet were able to recognise that it was a reducing sugar without justifying their answer. In weaker responses, candidates incorrectly identified the structure as a dipeptide, polymers, cellulose or a carbohydrate.
- b. i. Better responses clearly drew a structural equation representing the formation of a dipeptide with water also being produced. Weaker responses confused the structure of amino acids with that of carbohydrates. Weaker responses did not draw the correct structure of an amino acid, frequently using NH_3 . Many candidates did not write a full equation, omitting water as a product.
- ii. In better responses, candidates explained clearly the principles of paper chromatography, relating their answer to amino acids. Weaker responses confused proteins and amino acids, not recognising that amino acids are colourless. Weaker responses did not explain the separation of movement of amino acids in terms of polarity of the mobile and stationary phases.
- c. i. Better responses clearly and concisely described the analysis of DNA and linked that analysis to identifying relationships through the familial uniqueness of DNA. Weaker responses confused the steps of DNA analysis or described other forensic techniques.
- ii. Most candidates identified the fingerprint of the child.
- d. i. In better responses, candidates clearly explained the correct tests that could be carried out in the school laboratory, including their positive results leading to identification. Many candidates found the chemistry of this question challenging. In poorer responses, candidates confused the tests which could lead to elimination. Candidates frequently confused the tests for proteins and amino acids. Tests for acetylsalicylic acid and identification of potassium chloride appeared to be better known by candidates. Sucrose was generally identified by elimination, although better responses addressed hydrolyse then testing with Benedict's solution.
- ii. Better responses identified destructive tests and non-destructive tests. Poorer responses did not distinguish between the two types of tests, frequently naming tests unrelated to the question.

e. In better responses, candidates demonstrated a thorough knowledge of three techniques providing an evaluation of their best use in forensics.

In poorer responses, many candidates included Atomic Absorption Spectroscopy. Weaker responses did not fully address the question; by not describing the principles of the instrumental techniques or naming their use without an evaluation of that use.